Phycologia (2001) Volume 40 (3), 268–274 Published 13 August 2001

Nodularin analyses and toxicity of a *Nodularia spumigena* (Nostocales, Cyanobacteria) water-bloom in the western Gulf of Finland, Baltic Sea, in August 1999

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H.T. KANKAANPÄÄ, V.O. SIPIÄ, J.S. KUPARINEN, J.L. OTT AND W.W. CARMICHAEL. 2001. Nodularin analyses and toxicity of a *Nodularia spumigena* (Nostocales, Cyanobacteria) water-bloom in the western Gulf of Finland, Baltic Sea, in August 1999. *Phycologia* 40: 268–274.

During 2–12 August 1999, the development of a surface bloom dominated by the cyanobacterium *Nodularia spumigena* was followed in the western Gulf of Finland, Baltic Sea. Intense surface scums of aggregated *N. spumigena* existed on calm days. The low-nutrient *N. spumigena*-dominated water mass was separated by a front from a nutrient-rich water mass dominated by *Aphanizomenon* cf. *flos-aquae* (Nostocales, Cyanobacteria). Owing to inadequate mixing of the two water masses, it is likely that *N. spumigena* growth was based mainly on the use of internal phosphorus stores. Signs of cell decay were observed on 5 August, and the proportion of empty filaments in the community increased towards the end of the survey period. A high number of *Nitzschia* spp. (Bacillariophyceae) was recorded within the *N. spumigena* aggregates on 8 August, suggesting nutrient leakage from the *N. spumigena* filaments. Senescence of *N. spumigena* and a decrease in the proportion of intact *N. spumigena* cells in dry material resulted in decreasing concentrations of cell-bound nodularin (NODLN), from 2.1 to 0.5 g kg^{-1} dry weight. No clear temporal trends in NODLN concentrations (< 0.5–2.6 μg l^{-1}) were observed in Gulf waters. We hypothesize that the bloom was linked to a fish kill that occurred during the same time period. Threespine sticklebacks (*Gasterosteus aculeatus*, Gasterosteiformes, Gasterosteidae) were found floating on the surface during the water-bloom. The sticklebacks contained approximately 35–170 μg NODLN kg^{-1} dry weight (microcystin-LR equivalents). Empty *N. spumigena* cells and trace amounts of NODLN were also found in surficial sediment, indicating that toxin-producing *N. spumigena* had reached the seafloor.

INTRODUCTION

The Baltic Sea is a semi-enclosed, severely eutrophicated brackish sea with seasonal cyanobacterial blooms. Nitrogen and phosphorus, originating from land-based sources and internal loading from anoxic sediments are readily available in the Baltic Sea. Except in enclosed bays and narrow coastal areas close to point sources of pollution, annual primary production is regulated mostly by nitrogen. During the summer periods, when both N and P macronutrients are close to zero in the upper mixed layer, weather conditions and hydrodynamic factors play a key role in triggering cyanobacterial blooms (Kononen *et al.* 1996) and bringing phosphorus to surface waters.

Massive blooms of N_2-fixing cyanobacteria, including *Nodularia spumigena* Mertens *ex* Bornet & Flahault, *Aphanizomenon flos-aquae* (Linnaeus) Ralfs *ex* Bornet & Flahault and *Anabaena* Bory *ex* Bornet & Flahault spp., are common in late summer and limit the use of water for recreation. These blooms are frequently toxic owing to nodularin produced by certain *Nodularia* strains (Sivonen *et al.* 1989a). Blooms are also suspected of contributing organic matter that fuels large segments of bottom water hypoxia and anoxia throughout the main basins of the Baltic Sea.

Nodularia spumigena is a nitrogen-fixing cyanobacterium, which forms toxic blooms during the late summer (Sivonen *et al.* 1989a, b; Kononen *et al.* 1993). It produces a cyclic pentapeptide hepatotoxin called nodularin (NODLN; Rinehart *et al.* 1988). There are several NODLN analogues (Namikoshi *et al.* 1994), but in this paper NODLN refers to cyclo[-D-*erythro*-β-methylAsp(iso-linkage)-L-Arg-Adda-D-Glu(iso-linkage)-*N*-methyldehydrobutyrine], a water-soluble, cyclic pentapeptide toxin, which is the only reported NODLN analogue produced by the Baltic *N. spumigena* (Sivonen *et al.* 1989a).

Baltic Sea NODLN concentrations range from < 0.1–2.4 g kg^{-1} dry weight (dw) of water-bloom (Sivonen *et al.* 1989a) up to 18.1 g kg^{-1} dw (Kononen *et al.* 1993). NODLN intoxication causes diarrhoea, vomiting, weakness, anorexia, pallor of mucous membranes, and cold extremities (Carmichael 1997). NODLN is an inhibitor of serine–threonine phosphatases 1, 2A and 3 (Yoshizawa *et al.* 1990; Honkanen *et al.* 1994). In the Baltic Sea area, several animal poisonings have occurred (Kononen & Elbrächter 1996).

The acute LD_{50} of NODLN to mice (intraperitoneally, i.p.) is approximately 50 μg kg^{-1} (Carmichael *et al.* 1988). Long-term exposure to NODLN may promote tumours and induce liver cancer (Ohta *et al.* 1994) and apoptosis (Fladmark *et al.* 1998). Bioaccumulation of cyanobacterial toxins in fish and mussels is a health issue and a matter of economic importance. Microcystins (MCs), cyclic heptapeptides closely related to NODLN, are persistent in food webs (Sivonen & Jones 1999), but there is little information about the half-life of these toxins or of NODLN in natural biotas.

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Fig. 1. Study area in the western Gulf of Finland, the Baltic Sea.

In this paper, we focus on a *N. spumigena* bloom in the northern Baltic Sea that occurred in early August 1999. We report the changes of cell-bound NODLN concentrations during growth and senescence of this Baltic *N. spumigena* bloom and NODLN concentrations in threespine sticklebacks (*Gasterosteus aculeatus* Linnaeus) affected by the bloom.

MATERIAL AND METHODS

Sampling

The study site at the entrance to the Gulf of Finland is characterized by hydrodynamic activity caused by the inflow–outflow pattern (Alenius *et al.* 1998) and irregular bottom topography, resulting in the formation of fronts, upwelling, downwelling, jets and eddies (Haapala 1994). The R/V *Aranda* of the Finnish Institute of Marine Research (FIMR) was used as the main sampling platform. The study area and sampling information are presented in Figs 1, 2. All coordinates are in WGS-84 datum. At the main sampling station, LL12 (59°29.03′N; 22°53.82′E; 81 m water depth), daily surveys with at least two samplings per day were carried out during 2–5 and 8–11 August 1999. Water samples from 0–5 m depth (Fig. 2) were collected with Hydro-Bios (Germany) samplers, as well as CTD Rosette bottles.

Algae (2–5 and 8 August) and threespine stickleback (3, 4 and 8 August) samples were collected within a 1.5 km radius of the monitoring station LL12. Hauls of algae were made with a 50 μm plankton net from both the research vessel and motorboat. Net hauls were combined to produce pooled samples. Fish were collected from the motorboat, either by hand or by using fish net, within an approximately 1 km² area. Fish that were dead or in poor condition were observed on 3 August (20 individuals), 4 August (seven individuals) and 8 August (nine individuals), producing three sets of sticklebacks, weighing 13.3, 10.5 and 11.3 g, respectively. The position of the motorboat was determined by locking its position onto the ship's radar screen. A 40 cm long sediment sample was col-

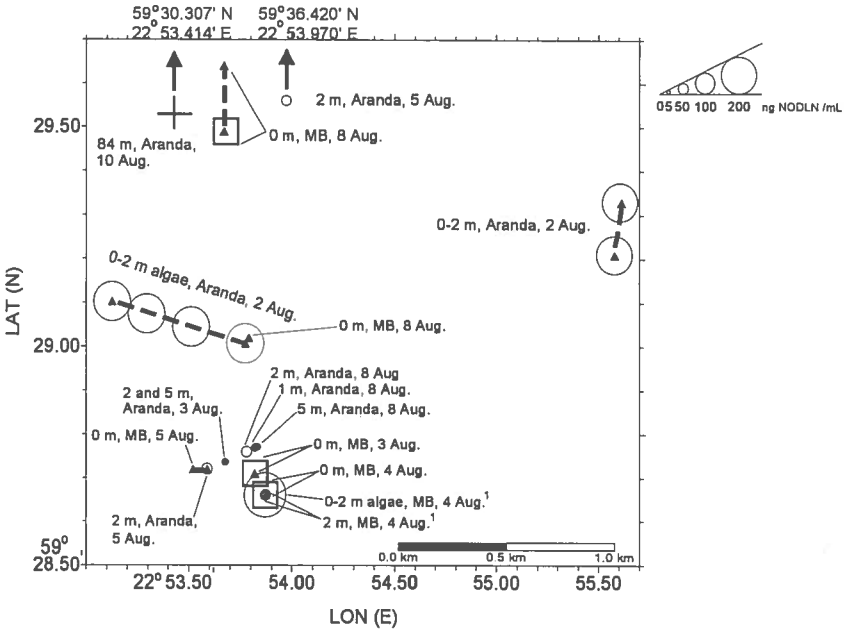

Fig. 2. Water sampling locations and dissolved NODLN concentrations (open circles; black circles = NODLN not detected) within the LL12 area during 2–8 August 1999, together with algae (triangles), fish (squares) and sediment (cross) sampling during 2–10 August 1999. Combined plankton supernatant samples, sampling depths, platforms and sampling dates are indicated. [1] 200 × 200 m survey area. MB = motorboat.

lected (84 m) on 10 August 1999, using a Gemini Twin corer. The uppermost 1 cm of the recent sediment was sliced for microscopy and NODLN analyses.

Solvents, reagents and standards

All solvents were high performance liquid chromatographic (HPLC) grade and reagents *pro analysi* grade. Deionized water was used, prepared with Millipore Milli-Q plus system. NODLN and microcystin-LR (MC-LR) were purchased from Calbiochem (La Jolla, CA, USA).

Salinity, temperature, fluorescence, oxygen and phosphate

Vertical temperature, salinity, fluorescence and oxygen profiles were obtained from the SeaBird 911Plus CTD profiler, equipped with a Sea Tech. fluorometer, and SBE 13 dissolved oxygen sensor (Sea-Bird Electronic, Bellevue, WA, USA). Nutrient samples were taken from the Rosette sampler mounted on the CTD. Dissolved inorganic phosphate was determined using a Lachat QuikChem FIA+ analyser (Zellweger Analytics, Milwaukee, WI, USA). Spatial and temporal changes in hydrography and phosphate were followed by night-time surveying of the study area. During the surveys (2–5 and 8–11 August), the water column was profiled at 20 locations.

Size fractionation for chlorophyll *a* (Chl *a*) was made by pouring the sample through a 20 μm mesh. The Chl *a* concentration was determined on Whatman GF/F filters after extraction at room temperature in the dark with 96% ethanol for 24 h. The concentration was measured with a Perkin Elmer LS-2B fluorometer.

Phytoplankton species were quantified from 500 ml samples preserved with Lugol's solution and counted with the Utermöhl (1958) technique. Cyanobacterial filaments were counted as 100 μm segments and other phytoplankton as single cells or colonies. The carbon biomass for species was calculated according to Edler (1979).

Sample pretreatment and extraction

Algae samples were lyophilized (Christ Alpha 1–4 Freeze Dryer, Frankfurt, Germany) for 24 h and ground with a mortar. Sticklebacks were washed thoroughly with tap water and deionized water and frozen. The viscera were separated and lyophilized (Edwards 12 K Super Modulyo Freeze Drier, West Sussex, England) for 1–3 days, homogenized (Fritsch Planetary Mill Pulverisette 5, Idar-Oberstein, Germany) and stored at −20°C. Particles were removed from water samples by centrifuging and filtration prior to analysis, but samples were not otherwise treated.

Dried algae (each 0.2 ± 0.01 g) and dried sticklebacks (pooled viscera samples of 0.05–0.2 g) were extracted with 20 ml (algae) or 3–5 ml (sticklebacks) of methanol for 8 h, using a Bransonic Branson 3200 ultrasonic bath (Dansbury, CT, USA). Fish extracts were concentrated 9–11-fold using a rotary evaporator (50°C). The sediment sample was prepared by extracting 0.7 ml of wet surface sediment with 0.7 ml of methanol for 5 min with a vortex mixer, after which the supernatant was separated by centrifuging.

ELISA analysis

Before ELISA, methanolic algae and fish extracts were diluted 10-fold with deionized water. Further 1000-fold or 10,000-fold dilutions were made depending on the NODLN concentration. Water samples were either analysed directly or diluted 10-fold and 100-fold with deionized water before analysis. All samples were filtered through 0.45 μm filters before analysis.

On board ship, algae and water samples were analysed using the EnviroLogix Tube Kit (ET 022; EnviroLogix, Portland, ME, USA), calibrated with 0.5, 1.5 and 3.0 ng ml^{-1} NODLN solutions. This kit is a competitive ELISA and has a detection limit of 0.5 ng ml^{-1} for NODLN. For stickleback samples, ELISA was performed at the Finnish Environment Institute using the EnviroGard® plate kit (Strategic Diagnostics, Newark, NJ, USA), with three-point MC-LR calibration. Pure NODLN dissolved in water caused 50% inhibition at 0.31 ng ml^{-1} and MC-LR at 0.37 ng ml^{-1}. Absorbance at 450 nm was measured with both ELISA kits. Fish results are expressed as MC-LR equivalents and samples of algae in NODLN equivalents.

HPLC analysis

On board ship, analysis of algae and sediment extracts was carried out with a Waters (Milford, MA, USA) HPLC system equipped with a Waters Sentry C_{18} (5 μm; 3.9 × 20 mm) precolumn and a Waters Symmetry C_{18} (3.9 × 150 mm) analytical column, and an isocratic mobile phase mixture of 0.01 M $(NH_4)_2HPO_4$, pH 2.0 (705 ml), and acetonitrile (295 ml). The flow rate was 1.7 ml min^{-1}. Injections of 1.0 μl (algae) and 10 μl (standards and sediment extract) were made using a Waters 717 autosampler. The eluate was monitored at 238 nm and at 190–300 nm using a Waters 996 photodiode array detector controlled by Waters Millennium[32] Chromatography Manager software. NODLN in the samples was identified by a comparison of the retention time (4.8 min) and UV spectrum with those of the NODLN standard (1–16 ng per 10 μl injection). The detection limit of the HPLC method was 0.3 ng standard NODLN per 10 μl injection (S/N ratio 18 ± 18%, 1.6 ng per 10 μl injection).

RESULTS AND DISCUSSION

Overall situation in the Gulf of Finland

An upwelling had taken place near to the Finnish coast 1 week before our cruise in August 1999. The upwelling brought cold, phosphorus-rich water to the surface layer, which suggested favourable growth conditions for cyanobacteria. Traces of the upwelling event were recorded in the northern part of our survey area in temperature isopleths. A less saline, cold water mass dominated by *A. flos-aquae* in the northern part of the survey area was separated by a moderate front from more saline, warm water (up to 20–21°C during 2–5 August). This southern area was dominated by *N. spumigena*. At the start of the survey, this front was crossing from NW to SE, but later it turned more parallel to the coast and moved towards the west, during the initial survey period. The aggregation process was weaker in the colder, northern water mass than in the warmer, *N. spumigena*-dominated water mass. A cross-

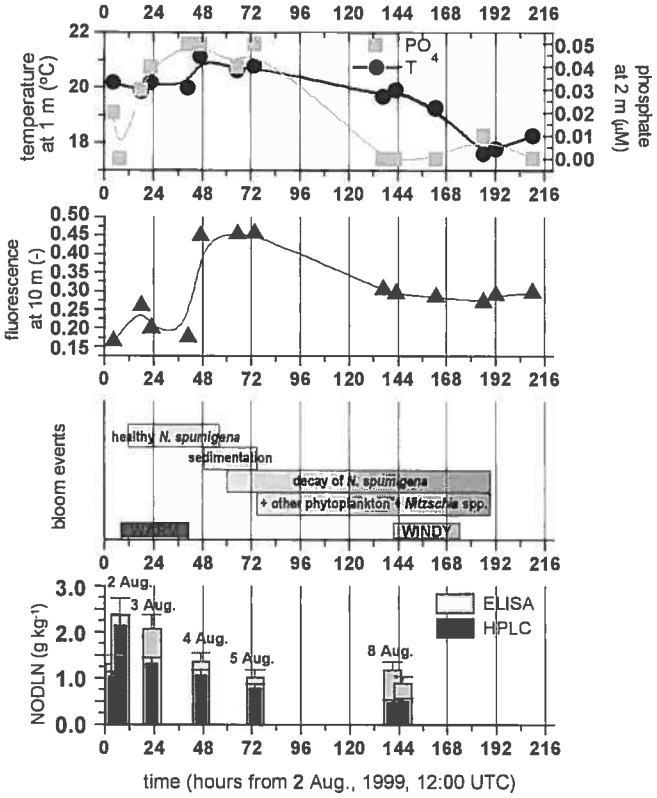

Fig. 3. Surface water temperature and phosphate concentrations, CTD fluorescence at 10 m, main events affecting the bloom and NODLN concentrations (ELISA: g kg^{-1}; ± 15%, HPLC: g kg^{-1}; ± 10%) in phytoplankton during 2–11 August 1999.

section from the survey area clearly indicated the different water masses and the upwelled water. Owing to the low salinity of the upwelled water, mixing with the *N. spumigena*-dominated, warmer water was not efficient and thus *N. spumigena* did not benefit from this upwelling.

On the southern, more saline side of the front, *N. spumigena* dominated the surface community and formed extensive centimetre sized aggregates. The survey area was covered with strips of surface accumulations several hundreds of metres long and 10–50 m wide. The water-bloom reached its peak during the calm period of 3–5 August. During this time, the phosphorus levels reached a maximum concentration of 0.05 µM, and then rapidly decreased to zero (Fig. 3).

The main thermocline was situated at a depth of 20–30 m, and there was another weaker thermocline near the surface. A clear oxygen minimum (4.7–5.9 ml l^{-1}) was observed just above the thermocline at 10–20 m, most probably owing to respiration of accumulated heterotrophic plankton on top of the thermocline. An autotrophic phytoplankton maximum (chlorophyll fluorescence) was situated at 4–10 m. This structure for the water column remained the same at station LL12 throughout 2–5 August, with only slightly fluctuating positions of the oxygen minimum and fluorescence maximum, brought about by changes in wind condition. Changes in 10 m Chl *a* fluorescence indicated sedimentation of phytoplankton during 4–5 August (Fig. 3). The net hauls revealed a slight increase in the amount of *N. spumigena* aggregates during this survey period.

Based on nutrient data from this cruise and previous (FIMR, unpublished observations) R/V *Aranda* cruises to the western Gulf of Finland, we hypothesize that the growth of *N. spumigena* was supported mainly by internal phosphorus stores from previous upwellings.

Table 1. Chlorophyll *a* (Chl *a*) concentrations in total plankton and nanoplankton (< 20 µm) at LL12 during the cruise.

	2–5 Aug.		8–11 Aug.	
Period	Total Chl *a* (mg m^{-3})	< 20 µm Chl *a* (mg m^{-3})	Total Chl *a* (mg m^{-3})	< 20 µm Chl *a* (mg m^{-3})
Depth(m)				
2	3.7 ± 0.7	2.6 ± 0.2	4.0 ± 0.6	3.9 ± 0.5
5	4.5 ± 0.9	3.2 ± 0.6	4.4 ± 0.6	3.7 ± 0.7

During 2–3 August, the phytoplankton community was clearly autotrophic and showed healthy growth. Microscopical counts showed that cyanobacteria dominated the surface (0–2 m) community, with less than 10% of the biomass being contributed by other algae (Dinophyceae, Bacillariophyceae and Chlorophyceae). *Nodularia spumigena* clearly dominated over *A. flos-aquae*, with a biomass ratio of 194 to 73 mg m^{-3}. The third dominant in the community was also a cyanobacterium, *Anabaena lemmermannii* P. Richter, but its biomass (15 mg m^{-3}) was less than 10% that of *N. spumigena*. Plankton net hauls from 20 m to the surface showed that *N. spumigena* dominated the whole surface water column, with a ratio of 5 : 2 in dominance over *A. flos-aquae*.

During 4–5 August, the Chl *a* maximum moved from the surface layer to 10 m depth (Fig. 3), suggesting light inhibition in the surface water or sedimentation of the plankton community caused by calm weather. During this period, oxygen uptake and production measurements from aggregates showed a shift from autotrophy to heterotrophy, with increasing bacterioplankton growth and accumulation into the aggregates. A moderate change in the plankton community on 5 August was recorded also by net hauls from 20 m to the surface, with a dominance of *N. spumigena* over *A. flos-aquae* in a ratio of 5 : 3. At 5 m depth, *A. flos-aquae* biomass values were higher than *N. spumigena* by 415–337 mg m^{-3}, suggesting that surface floating aggregates of *N. spumigena* were not in a healthy phase of growth, compared to *A. flos-aquae*.

Thus, during 2–5 August, the *N. spumigena*-dominated plankton community reached its maximum growth and biomass level, supported by the internal nutrient storage from previous upwelling events. Signs of the decay of *N. spumigena* were already evident on 5 August, but decay became more evident on our return to the station on 8 August. Microscopical observations showed clear decay of *N. spumigena* filaments during the end of the cruise. The appearance of healthy growths of species of the diatom *Nitzschia* Hassall among *N. spumigena* aggregates suggests leakage of nutrients from *N. spumigena* filaments. Also, biomass showed decreasing values towards the end of the cruise: the biomass of *N. spumigena* at 5 m depth decreased from 337 to 160 mg m^{-3} between 5 and 11 August, while *A. flos-aquae* biomass decreased from 415 to 149 mg m^{-3} during the same period.

Chlorophyll *a* values (Table 1) at LL12 showed the development of the phytoplankton community during the cruise. The total community Chl *a* level at 2 and 5 m depths remained at the same level throughout the cruise, but the contribution

Table 2. Nodularin levels in threespine stickleback viscera.

Date	No. of individuals	Toxin (μg kg^{-1} dw)[1]
3 Aug.	20	170
4 Aug.	7	35
8 Aug.	9	150

[1] Hepatotoxin concentration in pooled sample in microcystin-LR equivalents, precision ± 15%.

of nanoplankton (< 20 μm) increased towards the end of the cruise, especially at 2 m. The Chl *a* data support microscopical observations of the decaying filamentous cyanobacteria and the increased contribution of other algae in the community by the end of the cruise.

Analysis of NODLN

HPLC is widely used in the analysis of cyanobacterial hepatotoxins (Meriluoto 1997), as are nonchromatographic methods such as ELISA and PP inhibition assay (An & Carmichael 1994). Application of these methods for environmental monitoring of cyanotoxins has not been standardized between laboratories, but each has certain advantages. We found that HPLC and ELISA gave consistent results with good correlation (r = 0.92, n = 7). On average, ELISA values were 50% higher than HPLC for the alga samples (Fig. 3), which may be due to nonspecific binding in ELISA, or cyanobacterial hepatotoxins undetectable in HPLC. However, only one distinct peak (NODLN) was observed in HPLC.

Data from 12 three-point NODLN calibrations with the ELISA tube kit showed relative standard deviation values of 6–19%. Considering that the EnviroLogix tube kit is primarily meant for semiquantitative to qualitative field analysis, the kit's performance was acceptable for these applications. Toxin-free Norwegian flounder analysed previously (Sipiä *et al.* 2001) indicated undetectable (< 10 μg kg^{-1}) levels of cyanobacterial hepatotoxins. All three stickleback samples exceeded the blank tissue values and were thus positive for NODLN (Table 2). Recovery of NODLN from flounder liver spiked with NODLN standard (at 300 ng g^{-1}) was 29% (Sipiä *et al.* 2001). There are no data on recovery from sediments.

Development of phytoplankton and extracellular NODLN content

During 3–5 August, *N. spumigena* filaments appeared healthy. Cell ageing was evident after 8 August, when *Nitzschia* spp. cells were observed covering the *N. spumigena* filaments. Figure 3 illustrates the effect of cell ageing to the bloom, as reflected in statistically significant, decreasing NODLN cell concentrations (2.1–2.3 to 0.5–0.8 mg kg^{-1} dw). This can be explained by three factors: (1) sedimentation of phytoplankton containing *N. spumigena;* (2) biodilution of the algal samples by *A. flos-aquae, Nitzschia* spp. and other phytoplankton toward the end of the survey period; and (3) senescence of the *N. spumigena* cells. On 8 August, a heavy wind disrupted the bloom and senescence of the *N. spumigena* accelerated with decreasing cell-bound NODLN concentrations.

NODLN content in water varied between < 0.5 μg l^{-1} and 2.6 μg l^{-1} (measured by ELISA), with no temporal trends. Much higher NODLN concentrations (150–220 μg l^{-1}) were found in the supernatants of centrifuged plankton samples, indicating the presence of extracellular NODLN from *N. spumigena.* Most (> 80%) of the MCs or NODLN produced by healthy bloom populations are intracellular but, when waterblooms start to degrade, extracellular concentrations of 0.1–10 μg l^{-1} can be observed (Sivonen & Jones 1999). Park *et al.* (1998) have also reported elevated extracellular MC levels (> 20% of total) during cell senescence. Over 90% of MCs in lakes are degraded by bacterial activity within 2–10 days (Jones *et al.* 1994; Lahti *et al.* 1997). Studies conducted in Lakes Alexandrina and Albert in Australia indicated that *N. spumigena* cell numbers and NODLN water concentrations were strongly correlated, with a NODLN half-life of about 24 h (Heresztyn & Nicholson 1997).

Similarly, Twist & Codd (1997) observed that NODLN levels in aqueous solutions containing *N. spumigena* cells decreased within 9 days in laboratory conditions. Our results consist of cell-bound NODLN data from 5 days and show comparable NODLN behaviour. Although trends in water-phase concentrations were not seen, the results would suggest that NODLN rapidly disappears when the bloom is degraded.

Sedimentation of *N. spumigena* and NODLN

Empty *N. spumigena* cells were observed in the surface sediment, and a trace amount (*c.* 90 μg l^{-1}) of NODLN was found in the original sediment suspension from LL12. Thus, some NODLN may accumulate in sediments, at least temporarily. To date, accumulation of cyanobacterial toxins in sediments has gained little attention (Sivonen & Jones 1999). Sedimentation plays a key role in Baltic Sea mass balances and cyanobacteria constitute 14–88% of the late summer–autumn sedimentation, but only < 10% of total summer phytoplankton sedimentation (Kankaanpää *et al.* 1997a, b).

Baltic Sea cyanobacteria disintegrate mostly in the water column: < 1% of the daily cyanobacterial biomass was found to settle by Heiskanen (1998). Neither *N. spumigena* nor *A. flos-aquae* is effectively grazed by zooplankton (Haney 1987; Sellner 1997). Thus, disintegration of cyanobacteria is likely caused by bacteria and viruses. MC-LR is efficiently bound and removed from water by clay material (Morris *et al.* 2000), which may provide an effective way to remove MCs and probably NODLN from seawater and contribute to sedimentation of extracellular NODLN. Lysis of *N. spumigena* cells will probably continue on the sediment surface by the activity of benthic bacteria, but this has not been quantified for Baltic sediments. Bacteria capable of degrading cyanobacterial toxins, including NODLN, exist in lake sediments (Rapala *et al.* 1994).

Accumulation of NODLN in sticklebacks and health aspects

Cyanobacterial hepatotoxin (MC) bioaccumulation has been demonstrated in fish (Carbis *et al.* 1997; Sipiä *et al.* 2001). Unexplained seal mass mortalities in the Gulf of Finland occurred in 1991 and 1992 (Stenman & Westerling 1994), and bird mass-kills in 1991 and 2000; both may be connected to harmful algal blooms. MCs are also suspected to cause net-pen liver disease of farmed Atlantic salmon (Andersen *et al.* 1993). Using Zebra fish, Wiegand *et al.* (1999) observed that toxic effects of MC-LR involved the whole organism, which

may be due to the increased energy demand of the detoxification process.

Baltic threespine sticklebacks are important food to Baltic salmon, pike and some sea birds (Karlsson *et al.* 1999). The sticklebacks from the LL12 area were in poor shape or dead, and contained 35–170 μg NODLN g^{-1} dw (Table 2), thus indicating that they had ingested either *N. spumigena* cells or extracellular NODLN. Although we could not unambiguously assign the cause of death, it is probable that the *N. spumigena* bloom was linked to the fish kill. The stickleback results are based on total weight of viscera, not liver weight. Thus, direct comparison with earlier results is difficult. It is probable that NODLN concentrations in liver would exceed the values reported. Previously, we found NODLN in Baltic flounder and cod liver, in concentrations of 30–150 μg kg^{-1} dw (Sipiä *et al.* 2001), which is comparable with the present results. However, no previous results on acute effects have been reported.

No NODLN time-trends could be seen in sticklebacks, but it is noteworthy that the fish caught before the lysis of *N. spumigena* also contained appreciable amounts of NODLN. Anoxia is not a probable reason for the sticklebacks' poor condition, since the surface water always contained > 5 ml O_2 l^{-1}. The way NODLN entered the sticklebacks is uncertain, but since they also use phytoplankton as a food source, cell ingestion is possible. Because NODLN does not bind covalently to protein phosphatases (Bagu *et al.* 1995), it may be more available than MCs to the next trophic levels. Acute lethal human intoxication from algae or fish containing NODLN is unlikely. Orally administered NODLN affects liver about 200 times less effectively, compared with i.p. dosing (Kiviranta *et al.* 1990). Nevertheless, concern is not unjustified, and long-term effects to both biota and humans with subchronic levels should be considered.

Conclusions

This case study exemplified the importance of hydrodynamic control to *N. spumigena* bloom formation. Cell senescence was rapid and was reflected in a subsequent decrease in NODLN concentrations in phytoplankton. The increase in the number of *A. flos-aquae* and *Nitzschia* spp., and phytoplankton sedimentation from the surface layer, also contributed to the decrease. The tube ELISA kit was evaluated as a suitable tool for NODLN monitoring. And acute effects of both healthy and degrading *N. spumigena* bloom to sticklebacks were observed.

ACKNOWLEDGEMENTS

The authors express their gratitude to Ms Maija Huttunen for microscopic analyses and assistance in plankton sampling, Dr Kirsti Lahti (Finnish Environment Institute, Helsinki, Finland) for assistance in ELISA measurements, Mr Henry Söderman for help in sampling, Mr Tero Purokoski for CTD data and Mr Ilkka Lastumäki for nutrient analyses. We are also indebted to the crew of the R/V *Aranda* for invaluable assistance.

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Accepted 28 February 2001